Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

For unit vectors $\mathbf{b}$ and $\mathbf{c}$ and any non zero vector $\mathbf{a}$, the value of $\{\{(\mathbf{a}+\mathbf{b})\times(\mathbf{a}+\mathbf{c})\}\times(\mathbf{b}\times\mathbf{c})\}\cdot(\mathbf{b}+\mathbf{c})$ is

(A) $|\mathbf{a}|^2$

(B) $2|\mathbf{a}|^2$

(C) $3|\mathbf{a}|^2$

(D) none of these

Q.2

The vector $((\mathbf{i}-\mathbf{j})\times(\mathbf{j}-\mathbf{k}))\times(\mathbf{i}+5\mathbf{k})$ is equal to

(A) $5\mathbf{i}-4\mathbf{j}-\mathbf{k}$

(B) $3\mathbf{i}-2\mathbf{j}+5\mathbf{k}$

(C) $4\mathbf{i}-5\mathbf{j}-\mathbf{k}$

(D) $5\mathbf{i}+4\mathbf{j}-\mathbf{k}$

Q.3

If $\vec{a}=\hat{i}+\hat{j}+\hat{k},\vec{b}=4\hat{i}+3\hat{j}+4\hat{k}$, and $\vec{c}=\hat{i}+\alpha\hat{j}+\beta\hat{k}$ are linearly dependent vectors and $|\overrightarrow{c}|=\sqrt{3}$, then

(A) $\alpha=1,\beta=-1$

(B) $\alpha=1,\beta=\pm1$

(C) $\alpha=-1,\beta=\pm1$

(D) $\alpha=\pm1,\beta=1$

Q.4

Let, $\vec{a}=\vec{a}_1\hat{i}+\vec{a}_2\hat{j}+\vec{a}_3\hat{k},\vec{b}=\vec{b}_1\hat{i}+\vec{b}_2\hat{j}+\vec{b}_3\hat{k}\&$
$\vec{c}=\vec{c}_1\hat{i}+\vec{c}_2\hat{j}+\vec{c}_3\hat{k}$ be three non-zero vectors such that $\overrightarrow{c}$

@IITJEE_Advanced

is a unit vector perpendicular to both $\vec{a}\&\vec{b}$. If the angle between $\vec{a}\&\vec{b}$ is $\frac{\pi}{6}$, then $\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}^2$ is equal to

(A) 0

(B) 1

(C) $\frac{1}{4}|\vec{a}|^2|\vec{b}|^2$

(D) $\frac{3}{4}|\vec{a}|^2|\vec{b}|^2$

Q.5

Let $\vec{a}, \vec{b}\&\vec{c}$ be non-coplanar unit vectors equally inclined to one another at an acute angle θ. Then $\left|[\vec{a}\vec{b}\vec{c}]\right|$ in terms of θ is equal to

(A) $(1+\cos\theta)\sqrt{\cos 2\theta}$

(B) $(1+\cos\theta)\sqrt{1-2\cos 2\theta}$

(C) $(1-\cos\theta)\sqrt{1+2\cos\theta}$

(D) None of these

Q.6

If $\vec{a}$ and $\vec{b}$ are two non-zero non-collinear vectors then $2[\vec{a},\vec{b},\hat{i}]\hat{i}+2[\vec{a},\vec{b},\hat{j}]\hat{j}-2[\vec{a},\vec{b},\hat{k}]\hat{k}+[\vec{a},\vec{b},\vec{a}]$ is equal to

(A) $2(\vec{a}\times\vec{b})$

(B) $\vec{a}\times\vec{b}$

(C) $\vec{a}+\vec{b}$

(D) None of these

Q.7

Let $\vec{a}=\hat{i}-\hat{j}, \vec{b}=\hat{j}-\hat{k}$ and $\vec{c}=\hat{k}-\hat{i}$. If $\vec{d}$ is a unit vector such that $\vec{a}\cdot\vec{d}=0=[\vec{b}\vec{c}\vec{d}]$, then $\vec{d}$ equals

(A) $\pm\frac{\hat{i}+\hat{j}-2\hat{k}}{\sqrt{6}}$

(B) $\pm\frac{\hat{i}+\hat{j}-\hat{k}}{\sqrt{3}}$

(C) $\pm\frac{\hat{i}+\hat{j}+\hat{k}}{\sqrt{3}}$

(D) $\pm\hat{k}$

@IITJEE_Advanced

Q.8

If $\vec{r}\cdot\vec{a}=\vec{r}\cdot\vec{b}=\vec{r}\cdot\vec{c}=\frac{1}{2}$ for some non-zero vector $\vec{r}$, then the area of the triangle whose vertices are $A(\vec{a}), B(\vec{b})$ and $C(\vec{c})$ is ($\vec{a},\vec{b},\vec{c}$ are non-coplanar)

(A) $|[\vec{a}\vec{b}\vec{c}]|$

(B) $|\vec{r}|$

(C) $|[\vec{a}\vec{b}\vec{c}]\vec{r}|$

(D) None of these

Q.9

If $\vec{a},\vec{b},\vec{c}$ are three non-coplanar vectors, then the value of $\frac{\vec{a}\cdot(\vec{b}\times\vec{c})}{(\vec{c}\times\vec{a})\cdot\vec{b}}+\frac{\vec{b}\cdot(\vec{a}\times\vec{c})}{\vec{c}\cdot(\vec{a}\times\vec{b})}$ is :

(A) 0

(B) 2

(C) 1

(D) None of these

Q.10

Let a,b and c be three non-coplanar vectors, and let $\overrightarrow{\mathrm{p}},\overrightarrow{\mathrm{q}}$ and $\overrightarrow{\mathrm{r}}$ be the vectors defined by the relations $\overrightarrow{\mathrm{p}}=\frac{\vec{\mathrm{b}}\times\vec{\mathrm{c}}}{[\vec{\mathrm{a}}\,\vec{\mathrm{b}}\,\vec{\mathrm{c}}]},\overrightarrow{\mathrm{q}}=\frac{\vec{\mathrm{c}}\times\vec{\mathrm{a}}}{[\vec{\mathrm{a}}\,\vec{\mathrm{b}}\,\vec{\mathrm{c}}]}$ and $\overrightarrow{\mathrm{r}}=\frac{\vec{\mathrm{a}}\times\vec{\mathrm{b}}}{[\vec{\mathrm{a}}\,\vec{\mathrm{b}}\,\vec{\mathrm{c}}]}$. Then the value of the expression $(\vec{a}+\vec{b})\cdot\vec{p}+(\vec{b}+\vec{c})\cdot\vec{q}+(\vec{c}+\vec{a})\cdot\vec{r}$ is equal to

(A) 0

(B) 1

(C) 2

(D) 3

Q.11

If $\vec{u},\vec{v},\vec{w}$ are non-coplanar vectors and p,q are real numbers, then the equality $[3\vec{u}p\vec{v}p\vec{\omega}]-[p\vec{v}\vec{\omega}q\vec{u}]-[2\vec{\omega}q\vec{v}q\vec{u}]=0$ holds for:

(A) exactly two values of (p,q)

(B) more than two but not all values of (p,q)

@IITJEE_Advanced

(C) all values of (p, q)

(D) exactly one value of (p, q)

Q.12

Let $\vec{r} = (\vec{a} \times \vec{b}) \sin x + (\vec{b} \times \vec{c}) \cos y + 2(\vec{c} \times \vec{a})$
where $\vec{a}, \vec{b}, \vec{c}$ are three non-coplanar vectors.
If $\vec{r}$ is perpendicular to $\vec{a} + \vec{b} + \vec{c}$, then minimum value of $x^2 + y^2$ is

(A) π^2

(B) $\frac{\pi^2}{4}$

(C) $\frac{5\pi^2}{4}$

(D) None of these

Q.13

Let $\vec{a}, \vec{b}, \vec{c}$ be non-coplanar vectors and
$\overrightarrow{\mathrm{p}} = \frac{\overrightarrow{\mathrm{b}} \times \overrightarrow{\mathrm{c}}}{[\overrightarrow{\mathrm{a}}\,\overrightarrow{\mathrm{b}}\,\mathrm{c}]}, \overrightarrow{\mathrm{q}} = \frac{\overrightarrow{\mathrm{c}} \times \overrightarrow{\mathrm{a}}}{[\overrightarrow{\mathrm{abc}}]}, \overrightarrow{\mathrm{r}} = \frac{\overrightarrow{\mathrm{a}} \times \overrightarrow{\mathrm{b}}}{[\overrightarrow{\mathrm{abc}}]}$
What is the value of
$(\overrightarrow{\mathrm{a}} - \overrightarrow{\mathrm{b}} - \overrightarrow{\mathrm{c}}) \cdot \overrightarrow{\mathrm{p}} + (\overrightarrow{\mathrm{b}} - \overrightarrow{\mathrm{c}} - \overrightarrow{\mathrm{a}}) \cdot \overrightarrow{\mathrm{q}} + (\overrightarrow{\mathrm{c}} - \overrightarrow{\mathrm{a}} - \overrightarrow{\mathrm{b}}) \cdot \overrightarrow{\mathrm{r}}$?

(A) 0

(B) -3

(C) 3

(D) -9

Q.14

Let $\overrightarrow{\mathrm{A}} = \mathrm{a}_1\hat{\mathrm{i}} + \mathrm{a}_2\hat{\mathrm{j}} + \mathrm{a}_3\hat{\mathrm{k}}, \overrightarrow{\mathrm{B}} = \mathrm{b}_1\hat{\mathrm{i}} + \mathrm{b}_2\hat{\mathrm{j}} + \mathrm{b}_3\hat{\mathrm{k}}$ and $\overrightarrow{\mathrm{C}} = \mathrm{c}_1\hat{\mathrm{i}} + \mathrm{c}_2\hat{\mathrm{j}} + \mathrm{c}_3\hat{\mathrm{k}}$ be three non zero vectors such that $\overrightarrow{\mathrm{C}}$ is a unit vector perpendicular to both the vectors $\overrightarrow{\mathrm{A}}$ and $\overrightarrow{\mathrm{B}}$. If the angle between $\overrightarrow{\mathrm{A}}$ and $\overrightarrow{\mathrm{B}}$ is $\frac{\pi}{6}$, then $\begin{vmatrix} \mathrm{a}_1 & \mathrm{a}_2 & \mathrm{a}_3 \\ \mathrm{b}_1 & \mathrm{b}_2 & \mathrm{b}_3 \\ \mathrm{c}_1 & \mathrm{c}_2 & \mathrm{c}_3 \end{vmatrix}^2$ is equal to

(A) 0

(B) 1

(C) $\frac{1}{4}\left(a_1^2 + a_2^2 + a_3^2\right)\left(b_1^2 + b_2^2 + b_3^2\right)$

(D) $\frac{3}{4}\left(a_1^2 + a_2^2 + a_3^2\right)\left(b_1^2 + b_2^2 + b_3^2\right)\left(c_1^2 + c_2^2 + c_3^2\right)$

@IITJEE_Advanced

Q.15

Let $\mathbf{a}=\mathbf{i}-\mathbf{j}, \mathbf{b}=\mathbf{j}-\mathbf{k}, \mathbf{c}=\mathbf{k}-\mathbf{i}$. If $\hat{\mathrm{d}}$ is a unit vector such that $\mathbf{a.d}=0=[\mathrm{bc}\hat{\mathrm{d}}]$, then $\hat{\mathbf{d}}$ is equal to

(A) $\pm\frac{\mathbf{i}+\mathbf{j}-\mathbf{k}}{\sqrt{3}}$

(B) $\pm\frac{i+j+k}{\sqrt{3}}$

(C) $\pm\frac{i+j-2k}{\sqrt{6}}$

(D) $\pm\mathbf{k}$

Q.16

The volume of the tetrahedron whose vertices are with position vectors $\hat{\mathbf{i}}-6\hat{\mathbf{j}}+10\mathrm{k}, -\hat{\mathrm{i}}-3\hat{\mathbf{j}}+7\mathrm{k}, \hat{\mathrm{i}}-\hat{\mathrm{j}}+\lambda\mathrm{k}$ and $7\hat{\mathrm{i}}-4\hat{\mathrm{j}}+7\mathrm{k}$ is 11 cubic unit if λ equals

(A) -3

(B) 3

(C) 7

(D) -1

Q.17

If $(\vec{a}\times\vec{b})\times\vec{c}=\vec{a}\times(\vec{b}\times\vec{c})$ where $\vec{a},\vec{b}$ and $\vec{c}$ are any three vectors such that $\vec{a}\vec{b}\neq 0, \vec{b}\cdot\vec{c}\neq 0$ then $\vec{a}$ and $\vec{c}$ are

(A) inclined at an angle of $\frac{\pi}{3}$ between them

(B) inclined at an angle of $\frac{\pi}{6}$ between them

(C) perpendicular

(D) parallel

Q.18

The vectors $\vec{a},\vec{b},\vec{c}$ and $\vec{d}$ are such that $\vec{a}\times\vec{b}=\overrightarrow{\mathrm{c}}\times\mathrm{d}$ and $\overrightarrow{\mathrm{a}}\times\overrightarrow{\mathrm{c}}=\overrightarrow{\mathrm{b}}\times\overrightarrow{\mathrm{d}}$. Which of the following is/ are correct?

1. $(\overrightarrow{\mathrm{a}}-\overrightarrow{\mathrm{d}})\times(\overrightarrow{\mathrm{b}}-\overrightarrow{\mathrm{c}})=\overrightarrow{0}$
2. $(\overrightarrow{\mathrm{a}}\times\overrightarrow{\mathrm{b}})\times(\overrightarrow{\mathrm{c}}\times\overrightarrow{\mathrm{d}})=\overrightarrow{0}$

Select the correct answer using the code given below:

(A) 1 only

(B) 2 only

@IITJEE_Advanced

(C) Both 1 and 2

(D) Neither 1 nor 2

Q.19

$\hat{i} \times (\overrightarrow{A} \times \hat{i}) + \hat{j} \times (\overrightarrow{A} \times \hat{j}) + \hat{k} \times (\overrightarrow{A} \times \hat{k})$ is equal to

(A) $\overrightarrow{A}$

(B) $2\overrightarrow{A}$

(C) $3\overrightarrow{A}$

(D) 0

Q.20

$[(a \times b) \times (b \times c)(b \times c) \times (c \times a)(c \times a) \times (a \times b)] =$

(A) $[\mathrm{abc}]^2$

(B) $[abc]^3$

(C) $[abc]^4$

(D) None of these

Answers & Solutions

Q.1 Answer:

none of these

Solution:

The given expression

$$
\begin{aligned}
&= \{\{\mathbf{a} \times \mathbf{c} + \mathbf{b} \times \mathbf{a} + \mathbf{b} \times \mathbf{c}\} \times (\mathbf{b} \times \mathbf{c})\} \cdot (\mathbf{b} + \mathbf{c}) \\
&= \{(\mathbf{a} \times \mathbf{c}) \times (\mathbf{b} \times \mathbf{c}) + (\mathbf{b} \times \mathbf{a}) \times (\mathbf{b} \times \mathbf{c})\} \cdot (\mathbf{b} + \mathbf{c}) \\
&= [(\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}))\mathbf{c} - (\mathbf{c} \cdot (\mathbf{b} \times \mathbf{c}))\mathbf{a} + (\mathbf{b} \cdot (\mathbf{b} \times \mathbf{c}))\mathbf{a} \\
&-(\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})\mathbf{b})] \cdot (\mathbf{b} + \mathbf{c}) \\
&= [(\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}))(\mathbf{c} - \mathbf{b}) \cdot (\mathbf{b} + \mathbf{c})] \\
&= (\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}))\left[|\mathbf{c}|^2 - |\mathbf{b}|^2\right] = 0 \\
&\quad [\because |\mathbf{b}| = |\mathbf{c}| = 1].
\end{aligned}
$$

Q.2 Answer:

$5\mathbf{i} - 4\mathbf{j} - \mathbf{k}$

Solution:

The given expression

$((\mathbf{i} - \mathbf{j}) \cdot (\mathbf{i} + 5\mathbf{k}))(\mathbf{j} - \mathbf{k}) - ((\mathbf{j} - \mathbf{k}) \cdot (\mathbf{i} + 5\mathbf{k}))(\mathbf{i} - \mathbf{j})$

@IITJEE_Advanced

$= 1(\mathbf{j}-\mathbf{k})+5(\mathbf{i}-\mathbf{j}) = 5\mathbf{i}-4\mathbf{j}-\mathbf{k}.$

Q.3 Answer:

$\alpha = \pm 1, \beta = 1$

Solution:

If $\vec{a} = \hat{i}+\hat{j}+\hat{k}, \vec{b} = 4\hat{i}+3\hat{j}+4\hat{k}$ **and** $\vec{c} = \hat{i}+\alpha\hat{j}+\beta\hat{k}$

are linearly dependent vectors (provided) $|\overrightarrow{\mathbf{c}}| = \sqrt{3}$ **i.e.,** $[\vec{a}\vec{b}\vec{c}] = 0$

$$\Rightarrow \begin{vmatrix} 1 & 1 & 1 \\ 4 & 3 & 4 \\ 1 & \alpha & \beta \end{vmatrix} = 0 \Rightarrow \begin{vmatrix} 0 & 0 & 1 \\ 1 & -1 & 4 \\ 1-\alpha & \alpha-\beta & \beta \end{vmatrix} = 0$$

By expanding $\Rightarrow (\alpha-\beta-\alpha+1) = 0$

$1-\beta = 0 \Rightarrow \beta = 1$

Thus $|\vec{c}| = \sqrt{3} = \sqrt{1^2+\alpha^2+\beta^2} \Rightarrow \alpha = \pm 1$

Q.4 Answer:

$\frac{1}{4}|\vec{a}|^2|\vec{b}|^2$

Solution:

$$\because \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}^2 = [\vec{a}\vec{b}\vec{c}]^2$$

$= \{(\vec{a}\times\vec{b})\cdot\vec{c}\}^2 = |\vec{a}\times\vec{b}|^2|\vec{c}|^2\cos 0^\circ$

$[\because (\vec{a}\times\vec{b})$ **is parallel to** $\vec{c}]$

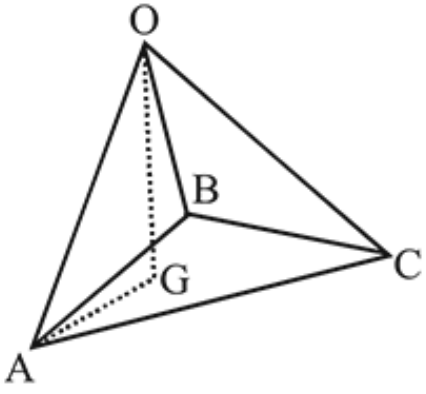
$= |\vec{a}|^2|\vec{b}|^2\sin^2\frac{\pi}{6}(\because |\vec{c}| = 1) = \frac{1}{4}|\vec{a}|^2|\vec{b}|^2$

Q.5 Answer:

$(1-\cos\theta)\sqrt{1+2\cos\theta}$

Solution:

$\overrightarrow{OA} = \overrightarrow{a}, \overrightarrow{OB} = \overrightarrow{b}$ & $\overrightarrow{OC} = \overrightarrow{c}$ **are unit vectors and equally inclined to each other at an acute angle** θ**.**



@IITJEE_Advanced

$\therefore ABC$ **is an equilateral triangle and**

$$AB = \sqrt{OA^2 + OB^2 - 2OA.OB \cdot \cos\theta}$$
$$= \sqrt{2 - 2\cos\theta} = \sqrt{2}\sqrt{1 - \cos\theta}$$

$\therefore$ **Area of** $\triangle ABC$

$$= \frac{\sqrt{3}}{4}AB^2 = \frac{\sqrt{3}}{4} \cdot 2(1 - \cos\theta) = \frac{\sqrt{3}}{2}(1 - \cos\theta)$$

If G **is the centroid of the** $\triangle ABC$**, then**

$$\alpha = \frac{1}{3}|\vec{a} + \vec{b} + \vec{c}|$$
$$\alpha = \frac{1}{3}\sqrt{a^2 + b^2 + c^2 + 2\vec{a} \cdot \vec{b} + 2\vec{b} \cdot \vec{c} + 2\vec{c} \cdot \vec{a}}$$
$$= \frac{1}{\sqrt{3}}\sqrt{1 + 2\cos\theta}$$

$$\therefore \left[\ \vec{a}\vec{b}\vec{c}\ \right] = \text{ Volume of parallelopipod}$$
$$= OG \times 2\,ar(\Delta ABC)$$
$$= 2 \cdot \frac{1}{\sqrt{3}}\sqrt{1 + 2\cos\theta} \times \frac{\sqrt{3}}{2}(1 - \cos\theta)$$
$$= (1 - \cos\theta)\sqrt{1 + 2\cos\theta}$$

Q.6 Answer:

$2(\vec{a} \times \vec{b})$

Solution:

$$2[\vec{a}, \vec{b}, \hat{i}]\hat{i} + 2[\vec{a}, \vec{b}, \hat{j}]\hat{j} - 2[\vec{a}, \vec{b}, \hat{k}]\hat{k} + [\vec{a}, \vec{b}, \vec{a}]$$
$$= 2(\vec{a} \times \vec{b})\hat{i}\hat{i} + 2(\vec{a} \times \vec{b})\hat{j} \cdot \hat{j} - 2(\vec{a} \times \vec{b})\hat{k} \cdot \hat{k} + 0$$
$$(\because [\vec{a}, \vec{a}, \vec{b}] = 0)$$
$$= 2(\vec{a} \times \vec{b}) + 2(\vec{a} \times \vec{b}) - 2(\vec{a} \times \vec{b})[\because \hat{i} \cdot \hat{i} = \hat{j}.\hat{j} = \hat{k} \cdot \hat{k} = 1]$$
$$= 2(\vec{a} \times \vec{b})$$

Q.7 Answer:

$\pm\frac{\hat{i}+\hat{j}-2\hat{k}}{\sqrt{6}}$

Solution:

Let $\vec{d} = x\hat{i} + y\hat{j} + z\hat{k}$

where $x^2 + y^2 + z^2 = 1$

$\therefore \vec{a} \cdot \vec{d} = 0 \Rightarrow x - y = 0$ **or** $x = y$

$$[\vec{b}\vec{c}\vec{d}] = 0 \Rightarrow \begin{vmatrix} 0 & 1 & -1 \\ -1 & 0 & 1 \\ x & y & z \end{vmatrix} = 0$$

or $x + y + z = 0$

or $2x + z = 0$**[Using (ii)] or** $z = -2x$ **From (i), (ii) and (iii), we have** $x^2 + x^2 + 4x^2 = 1$

@IITJEE_Advanced

$\Rightarrow x = \pm\frac{1}{\sqrt{6}}$

$\vec{d} = \pm\left(\frac{1}{\sqrt{6}}\hat{i} + \frac{1}{\sqrt{6}}\hat{j} - \frac{2}{\sqrt{6}}\vec{k}\right) = \pm\left(\frac{\hat{i}+\hat{j}-2\hat{k}}{\sqrt{6}}\right)$

Q.8 Answer:

$|[\vec{a}\vec{b}\vec{c}]\vec{r}|$

Solution:

Any vector $\vec{r}$ can be represented in terms of three non-coplanar vectors $\vec{a}, \vec{b}$ and $\vec{c}$ as $\vec{r} = x(\vec{a} \times \vec{b}) + y(\vec{b} \times \vec{c}) + z(\vec{c} \times \vec{a})$

Taking dot product with $\vec{a}, \vec{b}$ and $\vec{c}$, respectively, we have $x = \frac{\vec{r}\cdot\vec{c}}{[\vec{a}\vec{b}\vec{c}]}, y = \frac{\vec{r}\cdot\vec{a}}{[\vec{a}\vec{b}\vec{c}]}$ **and** $z = \frac{\vec{r}\cdot\vec{b}}{[\vec{a}\vec{b}\vec{c}]}$

From (i), we have

$[\vec{a}\vec{b}\vec{c}]\vec{r} = \frac{1}{2}(\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a})$

Area of $\Delta ABC = \frac{1}{2}|\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}|$

$= |[\vec{a}\vec{b}\vec{c}]\vec{r}|$

Q.9 Answer:

0

Solution:

By definition of scalar triple product $\vec{a} \cdot (\vec{b} \times \vec{c})$ can be written as $[\vec{a}\,\vec{b}\,\vec{c}]$

$\frac{\vec{a}\cdot(\vec{b}\times\vec{c})}{(\vec{c}\times\vec{a})\cdot\vec{b}} + \frac{\vec{b}\cdot(\vec{a}\times\vec{c})}{\vec{c}\cdot(\vec{a}\times\vec{b})} = \frac{[\vec{a}\vec{b}\vec{c}]}{[\vec{c}\vec{a}\vec{b}]} + \frac{[\vec{b}\vec{a}\vec{c}]}{[\vec{c}\vec{a}\vec{b}]}$

$= \frac{[\vec{a}\vec{b}\vec{c}]}{[\vec{a}\vec{b}\vec{c}]} - \frac{[\vec{a}\vec{b}\vec{c}]}{[\vec{a}\vec{b}\vec{c}]} = 1 - 1 = 0$

$\because [\vec{a}\vec{b}\vec{c}] = [\vec{b}\vec{c}\vec{a}] = [\vec{c}\vec{a}\vec{b}]$

but $[\vec{b}\vec{a}\vec{c}] = -[\vec{a}\vec{b}\vec{c}]$

Q.10 Answer:

3

Solution:

$\vec{a} \cdot \vec{p} = \frac{\vec{a}\cdot(\vec{b}\times\vec{c})}{[\vec{a}\vec{b}\vec{c}]} = \frac{[\vec{a}\vec{b}\vec{c}]}{[\vec{a}\vec{b}\vec{c}]} = 1 = \vec{b} \cdot \vec{q} = \vec{c} \cdot \vec{r}$

$\vec{b} \cdot \vec{p} = \frac{\vec{b}\cdot(\vec{b}\times\vec{c})}{[\vec{a}\vec{b}\vec{c}]} = \frac{0}{[\vec{a}\vec{b}\vec{c}]} = 0 = \vec{c} \cdot \vec{p} = \vec{a} \cdot \vec{r}$

Therefore, the given expression is equal to $1 + 0 + 1 + 0 + 1 + 0 = 3$

Q.11 Answer:

exactly one value of (p, q)

Solution:

$\because \vec{u}, \vec{v}, \vec{w}$ **are non-coplanar vectors**

$\therefore [\vec{u}, \vec{v}, \vec{w}] \neq 0$

@IITJEE_Advanced

Now, $[3\vec{u}, p\vec{v}, p\vec{w}] - [p\vec{v}, p\vec{w}, q\vec{u}] -$
$[2\vec{w}, q\vec{v}, q\vec{u}] = 0$
$\Rightarrow 3p^2[\vec{u}, \vec{v}, \vec{w}] - pq[\vec{v}, \vec{w}, \vec{u}] - 2q^2$
$[\vec{w}, \vec{v}, \vec{u}] = 0$
$\Rightarrow 3p^2[\vec{u}, \vec{v}, \vec{w}] - pq[\vec{u}, \vec{v}, \vec{w}] + 2q^2[\vec{u}, \vec{v}, \vec{w}]$
$\Rightarrow \left(3p^2 - pq + 2q^2\right)[\vec{u}, \vec{v}, \vec{w}] = 0$
$\Rightarrow 3p^2 - pq + 2q^2 = 0$
$\Rightarrow 2p^2 + p^2 - pq + \frac{q^2}{4} + \frac{7q^2}{4} = 0$
$\Rightarrow \quad 2\mathrm{p}^2 + \left(p - \frac{q}{2}\right)^2 + \frac{7}{4}q^2 = 0 \Rightarrow p = 0, q = 0$
$p = q/2$
This is possible only when $p = 0, q = 0 \therefore$ There is exactly one value of (p, q)

Q.12 Answer:

$\frac{5\pi^2}{4}$

Solution:

$\vec{r} = (\vec{a} \times \vec{b}) \sin x + (\vec{b} \times \vec{c}) \cos y + 2(\vec{c} \times \vec{a})$
$\vec{r} \cdot (\vec{a} + \vec{b} + \vec{c}) = 0$
$\Rightarrow [\vec{a}\vec{b}\vec{c}](\sin x + \cos y + 2) = 0$
Since, $[\vec{a}\vec{b}\vec{c}] \neq 0$, we have $\sin x + \cos y = -2$
This is possible only when $\sin x = -1$ and $\cos y = -1$.
For $x^2 + y^2$ to be minimum, $x = \frac{\pi}{2}$ and $y = \pi$.
$\therefore$ Minimum value of $\left(x^2 + y^2\right) = \frac{\pi^2}{4} + \pi^2 = \frac{5\pi^2}{4}$

Q.13 Answer:

3

Solution:

As given $\overrightarrow{\mathrm{p}} = \frac{\overrightarrow{\mathrm{b}} \times \overrightarrow{\mathrm{c}}}{[\vec{a}\vec{b}\vec{c}]}$, $\overrightarrow{\mathrm{q}} = \frac{\overrightarrow{\mathrm{c}} \times \overrightarrow{\mathrm{a}}}{[\mathrm{a}\,\overrightarrow{\mathrm{b}}\,\mathrm{c}]}$, and
$\overrightarrow{\mathrm{r}} = \frac{\overrightarrow{\mathrm{a}} \times \overrightarrow{\mathrm{b}}}{[\overrightarrow{\mathrm{a}}\,\overrightarrow{\mathrm{b}}\,\overrightarrow{\mathrm{c}}]}$
$\therefore \quad (\overrightarrow{\mathrm{a}} - \overrightarrow{\mathrm{b}} - \overrightarrow{\mathrm{c}}) \cdot \overrightarrow{\mathrm{p}} + (\overrightarrow{\mathrm{b}} - \overrightarrow{\mathrm{c}} - \overrightarrow{\mathrm{a}}) \cdot \overrightarrow{\mathrm{q}} + (\overrightarrow{\mathrm{c}} - \overrightarrow{\mathrm{a}} - \overrightarrow{\mathrm{b}}) \cdot \overrightarrow{\mathrm{r}}$
$= \frac{a \cdot (\vec{b} \times \vec{c})}{[\overrightarrow{abc}]} + \frac{\vec{b} \cdot (\vec{c} \times \vec{a})}{[\overrightarrow{abc}]} + \frac{\vec{c} \cdot (\vec{a} \times \vec{b})}{[\overrightarrow{abc}]}$
[**Since** $\overrightarrow{\mathrm{b}} \cdot (\overrightarrow{\mathrm{b}} \times \overrightarrow{\mathrm{c}}) = 0$
$\vec{c} \cdot (\overrightarrow{\mathrm{b}} \times \overrightarrow{\mathrm{c}}), = 0, \overrightarrow{\mathrm{c}} \cdot (\overrightarrow{\mathrm{c}} \times \overrightarrow{\mathrm{a}}) = 0 \overrightarrow{\mathrm{a}} \cdot (\overrightarrow{\mathrm{c}} \times \overrightarrow{\mathrm{a}})]$
$= 0, \vec{a} \cdot (\vec{a} \times \vec{b}) = 0$ **and** $\vec{b} \cdot (\vec{a} \times \vec{b}) = 0$
$= \frac{[\vec{a}\vec{b}\vec{c}]}{[\vec{a}\vec{b}\vec{c}]} + \frac{[\vec{a}\vec{b}\vec{c}]}{[\vec{a}\vec{b}\vec{c}]} + \frac{[\vec{a}\vec{b}\vec{c}]}{[\vec{a}\vec{b}\vec{c}]} = 3$

Q.14 Answer:

@IITJEE_Advanced

$\frac{1}{4}\left(a_1^2+a_2^2+a_3^2\right)\left(b_1^2+b_2^2+b_3^2\right)$

Solution:

$$\left|\begin{array}{ccc} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{array}\right|^2 = [\vec{A}\,\vec{B}\,\vec{C}]^2 = ((\vec{A}\times\vec{B})\cdot\vec{C})^2$$

$$= \left\{|\vec{A}||\vec{B}|\sin\tfrac{\pi}{6}(\vec{C})\cdot\vec{C}\right\}^2$$

$$= |\vec{A}|^2|\vec{B}|^2\left(\tfrac{1}{2}\right)^2|\vec{C}|^4 = \tfrac{1}{4}|\vec{A}|^2|\vec{B}|^2$$

$$= \tfrac{1}{4}\left(a_1^2+a_2^2+a_3^2\right)\left(b_1^2+b_2^2+b_3^2\right)$$

Q.15 Answer:

$\pm\frac{i+j-2k}{\sqrt{6}}$

Solution:

Let $\mathbf{a}=\mathbf{i}-\mathbf{j}, \mathbf{b}=\mathbf{j}-\mathbf{k}$ **and** $\mathbf{c}=\mathbf{k}-\mathbf{i}$

Let $\hat{d}=a_1 i+a_2 j+a_3 k, |\hat{d}|=\sqrt{a_1^2+a_2^2+a_3^2}=1$

$\Rightarrow a_1^2+a_2^2+a_3^2=1$

a. $\hat{d}=0 \Rightarrow a_1-a_2=0$

$[bca]=0 \Rightarrow b.(c\times d)=0$

$$\Rightarrow \left|\begin{array}{ccc} 0 & 1 & -1 \\ -1 & 0 & 1 \\ a_1 & a_2 & a_3 \end{array}\right| = -1\left(-a_3-a_1\right)-1\left(-a_2\right)$$

$\therefore a_1+a_2+a_3=0 \Rightarrow a_1-a_2+0a_3=0,$ **{ from (ii)\}**

$\therefore \frac{a_1}{0+1}=\frac{a_2}{1-0}=\frac{a_3}{-1-1} \Rightarrow \frac{a_1}{1}=\frac{a_2}{1}=\frac{a_3}{-2}=\lambda,$ **(say)**

$\Rightarrow a_1=\lambda, a_2=\lambda_3 a_3=-2\lambda$

$\therefore \lambda^2+\lambda^2+4\lambda^2=1,$ **from (i)\}**

$\Rightarrow 6\lambda^2=1 \Rightarrow \lambda=\pm\frac{1}{\sqrt{6}}; \therefore \hat{d}=\pm\frac{i+j-2k}{\sqrt{6}}$

Q.16 Answer:

7

Solution:

Let vertices are A, B, C, D **and** O **is origin**

$\therefore \overrightarrow{OA}=\hat{i}-6\hat{j}+10\hat{k} \quad \overrightarrow{OB}=-\hat{i}-3\hat{j}+7\hat{k}$

$\overrightarrow{CC}=5\hat{1}-\hat{j}+\lambda\hat{k}$

$\overrightarrow{OD}=7\hat{i}-4\hat{j}+7\hat{k}$

$\therefore \overrightarrow{AB}=\overrightarrow{OB}-\overrightarrow{OA}=-2\hat{i}+3\hat{j}-3\hat{k}$

$\overrightarrow{AC}=\overrightarrow{OC}-\overrightarrow{OA}=4\hat{i}+5\hat{j}+(\lambda-10)\hat{k}$

$\overrightarrow{AD}=\overrightarrow{OD}-\overrightarrow{OA}=6\hat{i}+2\hat{j}-3k$

Volume of tetrahedron

@IITJEE_Advanced

$= \frac{1}{6}[\overrightarrow{AB}, \overrightarrow{AC}, \overrightarrow{AD}] = \frac{1}{6}\begin{vmatrix} -2 & 3 & -3 \\ 4 & 5 & \lambda - 10 \\ 6 & 2 & -3 \end{vmatrix} = \frac{1}{6}\{-2(-15 - 2\lambda + 20) - 3(-12 - 6\lambda + 60) - 3(8 - 30)\}$

$= \frac{1}{6}\{4\lambda - 10 - 144 + 18\lambda + 66\}$

$= \frac{1}{6}(22\lambda - 88) = 11$ **(given)**

$\Rightarrow 2\lambda - 8 = 6$

$\therefore \lambda = 7$

Q.17 Answer:

parallel

Solution:

$(\vec{a} \times \vec{b}) \times \vec{c} = \vec{a} \times (\vec{b} \times \vec{c}), \vec{a} \cdot \vec{b} \neq 0$

$\vec{b} \cdot \vec{c} \neq 0$

$\Rightarrow (\vec{a} \cdot \vec{c}) \cdot \vec{b} - (\vec{b} \cdot \vec{c})\vec{a} = (\vec{a} \cdot \vec{c}) \cdot \vec{b} - (\vec{a} \cdot \vec{b}) \cdot \vec{c}$

$\Rightarrow (\vec{a} \cdot \vec{b}) \cdot \vec{c} = (\vec{b} \cdot \vec{c})\vec{a} \Rightarrow \vec{a} \| \vec{c}$

Q.18 Answer:

Both 1 and 2

Solution:

$(\overrightarrow{\mathrm{a}} - \overrightarrow{\mathrm{d}}) \times (\overrightarrow{\mathrm{b}} - \overrightarrow{\mathrm{c}})$

$= \overrightarrow{\mathrm{a}} \times \overrightarrow{\mathrm{b}} - \overrightarrow{\mathrm{d}} \times \overrightarrow{\mathrm{b}} - \overrightarrow{\mathrm{a}} \times \overrightarrow{\mathrm{c}} + \overrightarrow{\mathrm{d}} \times \overrightarrow{\mathrm{c}}$

$= \overrightarrow{\mathrm{c}} \times \overrightarrow{\mathrm{d}} - \overrightarrow{\mathrm{d}} \times \overrightarrow{\mathrm{b}} - \overrightarrow{\mathrm{b}} \times \overrightarrow{\mathrm{d}} - \overrightarrow{\mathrm{c}} \times \overrightarrow{\mathrm{d}}$

$= -\overrightarrow{\mathrm{d}} \times \overrightarrow{\mathrm{b}} + \overrightarrow{\mathrm{d}} \times \overrightarrow{\mathrm{b}}$

$= 0$

Again $(\vec{a} \times \vec{b}) = (\vec{c} \times \vec{d})$ **given**

$\Rightarrow (\overrightarrow{\mathrm{a}} \times \overrightarrow{\mathrm{b}}) \times (\overrightarrow{\mathrm{c}} \times \overrightarrow{\mathrm{d}}) = (\overrightarrow{\mathrm{c}} \times \overrightarrow{\mathrm{d}}) \times (\overrightarrow{\mathrm{c}} \times \overrightarrow{\mathrm{d}}) = 0$

(**as** $\overrightarrow{\mathrm{a}} \times \overrightarrow{\mathrm{a}} = 0$)

So both (1) and (2) are correct.

Q.19 Answer:

$2\overrightarrow{\mathrm{A}}$

Solution:

We have $\hat{i} \times (\overrightarrow{\mathrm{A}} \times \hat{i}) + \hat{j} \times (\overrightarrow{\mathrm{A}} \times \hat{j}) + \hat{k} \times (\overrightarrow{\mathrm{A}} \times \hat{k})$

$\hat{i} \times (\overrightarrow{\mathrm{A}} \times \hat{i}) = (\hat{i} \cdot \hat{i})\overrightarrow{\mathrm{A}} - (\hat{i} \cdot \overrightarrow{\mathrm{A}})\hat{i} = \overrightarrow{\mathrm{A}} - (\hat{i} \cdot \overrightarrow{\mathrm{A}})\hat{i} \ldots..(\mathrm{i})$

$\hat{j} \times (\overrightarrow{\mathrm{A}} \times \hat{j}) = (\hat{j} \cdot \hat{j})\overrightarrow{\mathrm{A}} - (\hat{j} \cdot \overrightarrow{\mathrm{A}})\hat{j} = \overrightarrow{\mathrm{A}} - (\hat{j} \cdot \overrightarrow{\mathrm{A}})\hat{j}$

And $\hat{k} \times (\overrightarrow{\mathrm{A}} \times \hat{k}) = (\hat{k} \cdot \hat{k})\overrightarrow{\mathrm{A}} - (\hat{k} \cdot \overrightarrow{\mathrm{A}})\hat{k}$

@IITJEE_Advanced

$= \overrightarrow{A} - (\hat{k} \cdot \overrightarrow{A})\hat{k}$

Now, eqn (i) + eqn (ii) + eqn (iii):

$\hat{i} \times (\overrightarrow{A} \times \hat{i}) + \hat{j} \times (\overrightarrow{A} \times \hat{j}) + \hat{k}(\overrightarrow{A} \times \hat{k}) = 3\overrightarrow{A}$

$-[(\hat{i} \cdot \overrightarrow{A})\hat{i} + (\hat{j} \cdot \overrightarrow{A})\hat{j} + (\hat{k} \cdot \overrightarrow{A})\hat{k}] = 3\overrightarrow{A} - \overrightarrow{A} = 2\overrightarrow{A}$

Q.20 Answer:

$[abc]^4$

Solution:

We have $(a \times b) \times (b \times c)$

$= ((a \times b) \cdot c)b - ((a \times b) \cdot b)c = [abc]b$

$(b \times c) \times (c \times a) = ((b \times c) \cdot a)c - ((b \times c).c)a = [bca]c$

$(c \times a) \times (a \times b) = ((c \times a), b)a - ((c \times a), a)b = [cab]a$

$\therefore [(a \times b) \times (b \times c)(b \times c) \times (c \times a)(c \times a) \times (a \times b)]$

$= [[abc]a[abc]b[abc]c] = [abc]^3[abc] = [abc]^4$

@IITJEE_Advanced